Physics 214/224, Quiz 5

Primary instructor: Yurii Maravin

04/26/2024

For grading purposes

Problem 1	
Problem 2	
Problem 3	
Problem 4	
Studio	
Total:	

Key

WID:____________________________ NAME:______________________________________

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

For two quick points, circle your studio instructor:

Lohman (TU 7:30) Maravin (TU 9:30) Maravin (TU 11:30)

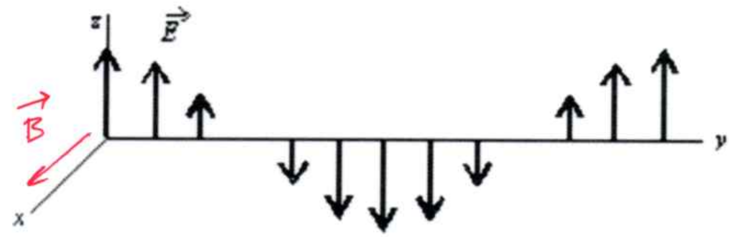
Han (WF 9:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.

You are almost done with EP2! You did a great job, good luck and ace this Quiz!

Part I: All questions to be answered on your Scantron card (48 points total)

1. (3 points) The EM wave moves in the positive direction of y-axis. The instantaneous snapshot of electric field vectors $\vec{E}$ are given in the figure below. Which way the magnetic field points at the origin?



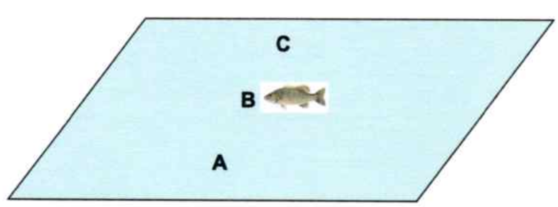
$\vec{c} \sim \vec{E} \times \vec{B}$

$\vec{c}$

   (a) Nowhere, it must be zero.
   (b) Along the positive direction of the x-axis.
   (c) Along the positive direction of the z-axis.
   (d) Along the negative direction of the x-axis.
   (e) Along the negative direction of the z-axis.

2. (3 points) Which property of a light wave *does not* change as the wave passes from one transparent medium to another with different indices of refraction?
   (a) Frequency.
   (b) Wavelength.
   (c) Velocity.
   (d) Amplitude.
   (e) Nothing is changing.

3. (3 points) Where should you aim your bow to shoot a smallmouth bass illustrated in the figure below?



   (a) A
   (b) B
   (c) C
   (d) It is not possible to shoot the fish with the bow, you need to have a hook, line, and fishing rod instead (and some bait).

4. (3 points) The most important characteristic of light that leads to the optical fiber's superior data transmission capability is?

   (a) Frequency.

   (b) Amplitude.

   (c) Polarization.

   (d) Intensity.

   (e) There is nothing superior in fiber optics. It is all government conspiracy or aliens.

5. (3 points) Unpolarized light with an intensity of 10.00 mW/m^2 passes through a single polarizer. The polarization axis of the polarizer is unknown. What is the intensity of the light upon exiting the polarizer?

   (a) 0.00 mW/m^2

   (b) 2.50 mW/m^2

   (c) 5.00 mW/m^2

   (d) 10.00 mW/m^2

   (e) We cannot identify the intensity of the light as we do not know the orientation of the polarization axis.

6. (3 points) A photometer is an instrument that measures the intensity of light at a particular location. For which type of image would a photometer register a non-zero intensity? ~~of a sun's image?~~

   due to the image

   (a) virtual image only

   (b) real or virtual image

   (c) real image only

   (d) neither; detectible light only exists at the object.

7. (3 points) To what distance should you focus your camera to take a picture of your image from a plane mirror located 3 m in front of you?

   (a) 3.00 m.

   (b) 0.33 m.

   (c) 4.00 m.

   (d) 6.00 m.

   (e) It is impossible to make a picture of yourself through the mirror as the image is virtual.

8. (3 points) Diverging lens can make

   (a) diverging images

   (b) real images

   (c) virtual images

   (d) can do both real and virtual images

9. (3 points) Concave mirror can make

   (a) diverging images
   (b) real images
   (c) virtual images
   (d) can do both real and virtual images

10. (3 points) Interference of light is an evidence that

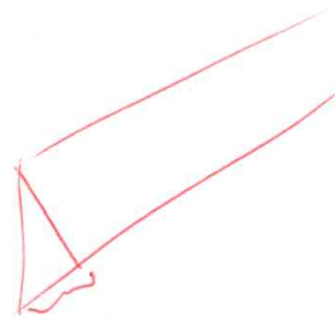
   (a) Light is electromagnetic in character.
   (b) The speed of light is very large.
   (c) Light does not obey the conservation of energy.
   (d) Light is a wave phenomenon.
   (e) Light is a transverse wave.

11. (3 points) What is the smallest angle an eye of a proctor with a pupil diameter of 5 mm could resolve for a $\lambda = 410$ nm light?

   (a) 100 mrad.
   (b) 10 mrad.
   (c) 1 mrad.
   (d) 0.1 mrad.
   (e) Proctor cannot see anything since he is asleep!

12. (3 points) Optical path length difference units are

   (a) Hz/m.
   (b) 1/m.
   (c) m/Hz.
   (d) m.
   (e) m/s.



13. (3 points) If two coherent beams of equal intensity blue light ($\lambda = 450$ nm) reach a point on a screen with the second beam traveling 900 nm farther than the first, the intensity of the light on the screen will be

   (a) zero.
   (b) the same as that of either of the two beams by itself.
   (c) twice that of either beam by itself.
   (d) four times that of either beam by itself.

14. (3 points) If two coherent beams of equal intensity red light ($\lambda = 670$ nm) reach a point on a screen with the second beam traveling 335 nm farther than the first, the intensity of the light on the screen will be

(a) zero.

(b) the same as that of either of the two beams by itself.

(c) twice that of either beam by itself.

(d) four times that of either beam by itself.

15. (3 points) Beams of green ($\lambda = 530$ nm) and red ($\lambda = 670$ nm) light are **both** passed through a 3 μm width slit and viewed on a second screen. What is the best description of what we would expect to see?

(a) The green and red light will be sharply concentrated only in the region of the viewing screen directly behind the slit.

(b) Both beams will form a central spot on the viewing screen that is spread out, with the red light spread more than the green light.

(c) Both beams will form a central spot on the viewing screen that is spread out, with the green light spread more than the red light.

(d) Both beams will spread out uniformly over the entire area of the viewing screen.

16. (3 points) Diffraction plays an important role in which of the following phenomena?

(a) The sun appears as a disk rather than a point to the naked eye. (Do not look at the sun without special protective glasses!!!)

(b) Light is bent as it passes through a glass prism.

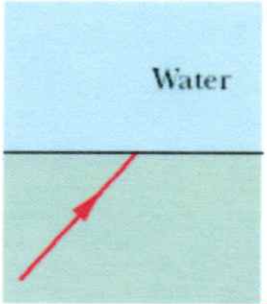
(c) A cheerleader yells through a megaphone.

(d) A farsighted person uses eyeglasses of positive focal length.

(e) A thin soap film exhibits colors when illuminated with white light.

Part II: Work out this part in the space provided. Show your work! (50 points total)

17. (10 points) In a figure below, a light ray in an underlying material is incident at angle $\theta_1 = 40°$ on a boundary with water, and some of the light reflects back into material and some refracts into the water. The index of refraction of the material is n_1 and that of water is $n_2 = 1.3$.



Give an expression that depends on n_1

(a) (3 points) What is the angle of refraction θ_2 (the answer will depend on n_1)?

(+2) $n_1 \sin\theta_1 = n_2 \sin\theta_2 \Rightarrow$

$$\theta_2 = \sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right) = \sin^{-1}\left(\frac{n_1 \sin 40°}{1.3}\right) \quad (+1)$$

(b) (3 points) If $n_1 > n_2$, is angle of refraction θ_2 bigger or smaller than θ_1?

Expl. (+2) if $n_1 > n_2 \Rightarrow \sin\theta_1 < \sin\theta_2$ from Snell's law $\Rightarrow$

$\theta_1 < \theta_2$ (θ_2 is bigger) (+1)

(c) (4 points) Assuming that $n_1 = 1.8$, at some range of incident angles θ_1, refraction into water is not possible and all light is reflected back into the material. Find this range of angles θ_1. (Note the word "range", do not just provide a single value).

From (a) $\theta_2 = \sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right)$ so at all times

Expl. (+2)

range (+1)

correct answer (+1)

$$\frac{n_1 \sin\theta_1}{n_2} \leq 1 \quad \text{or} \quad \sin\theta_1 \leq \frac{n_2}{n_1} = \frac{1.3}{1.8} \approx 0.72$$

This corresponds to $\theta_1 \leq 0.81$ rad or $\theta_1 \leq 46°$

For angles above 46°, the light will reflect back in the material, i.e. $\underline{46° \leq \theta_1 \leq 90°}$

18. (18 points) Impressed by the excellent performance of students in EP2 Quiz 5, Dr. Maravin decided to give extra points for extra rays in ray tracing problems 18 and 19. However, he noticed that he could not focus well on the grade book to read students' names. On that particular morning, he could focus only on the computer screen from a distance of 1.0 meter away!

(a) (2 point) What is Dr. Maravin's near point?

+2 $S_{NP} = 1\ m$ ← place where Dr. Maravin can still see a sharp image

(b) (4 points) Dr. Maravin wants to see clearly at a closer distance, so he puts on his contact lenses. With these contact lenses, he can see the gradebook clearly at a distance of 25 cm from his eyes. Where is the image that contact lenses create? Explain in words and provide the value of i.

Expl. +2 Image must be at $S_{NP} = 1\ m$, $i = -1\ m$, negative

Correct answer +2 value is due to the fact that the image is in the same hemisphere as the object

(c) (4 points) Determine the focal length of his contact lenses.

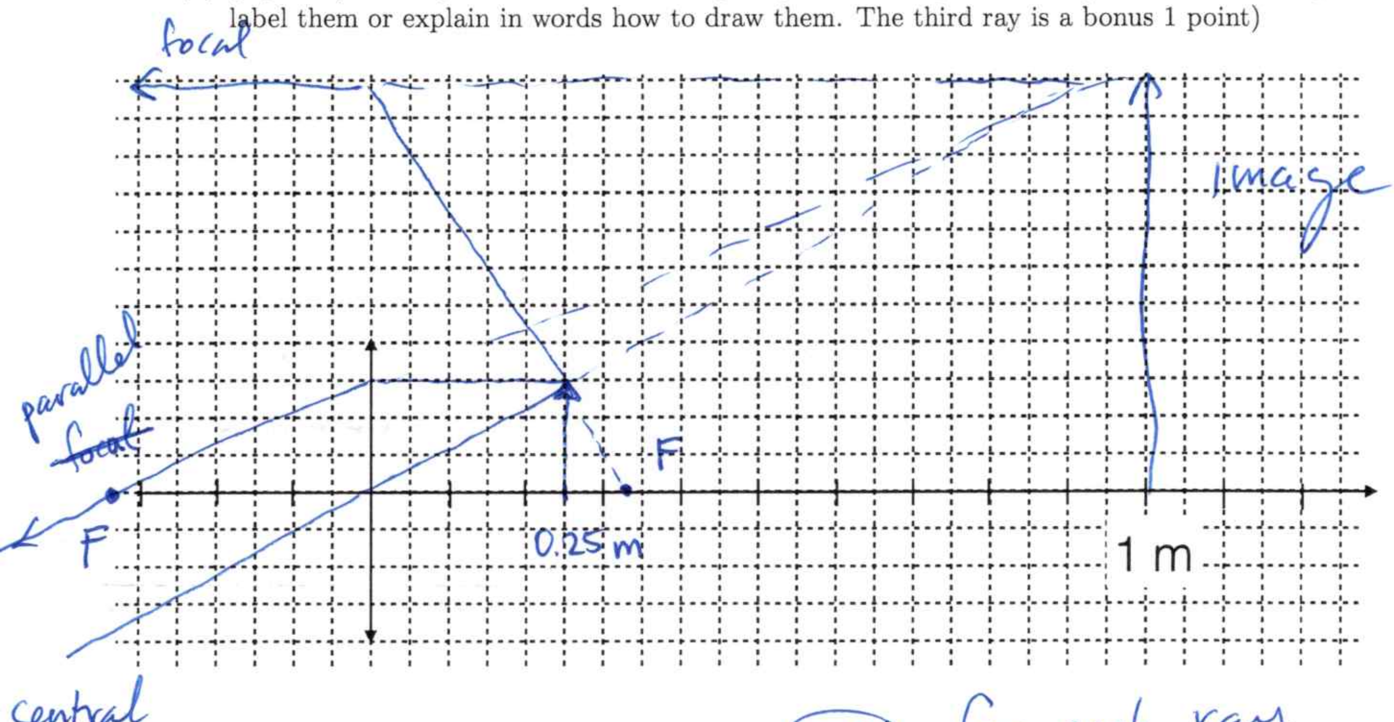
$$\frac{1}{p} + \frac{1}{i} = \frac{1}{f} \quad (+1) \qquad \frac{1}{0.25\,m} + \frac{1}{-1\,m} = \frac{1}{f} \quad (+2) \quad \Rightarrow \quad f \simeq 0.33\,m \quad (+1)$$

(d) (4 points) Is image virtual or real? Is it magnified or demagnified? (I will stress again, please explain your answer well, just writing an answer without explanation will get your points taken off).

As $i < 0$ – image is virtual +1 Expl. +1

$$\underbrace{m = -\frac{i}{p}}_{+1} = -\frac{-1\,m}{0.25} = 4 > 1 \Rightarrow \text{magnified} \quad (+1)$$

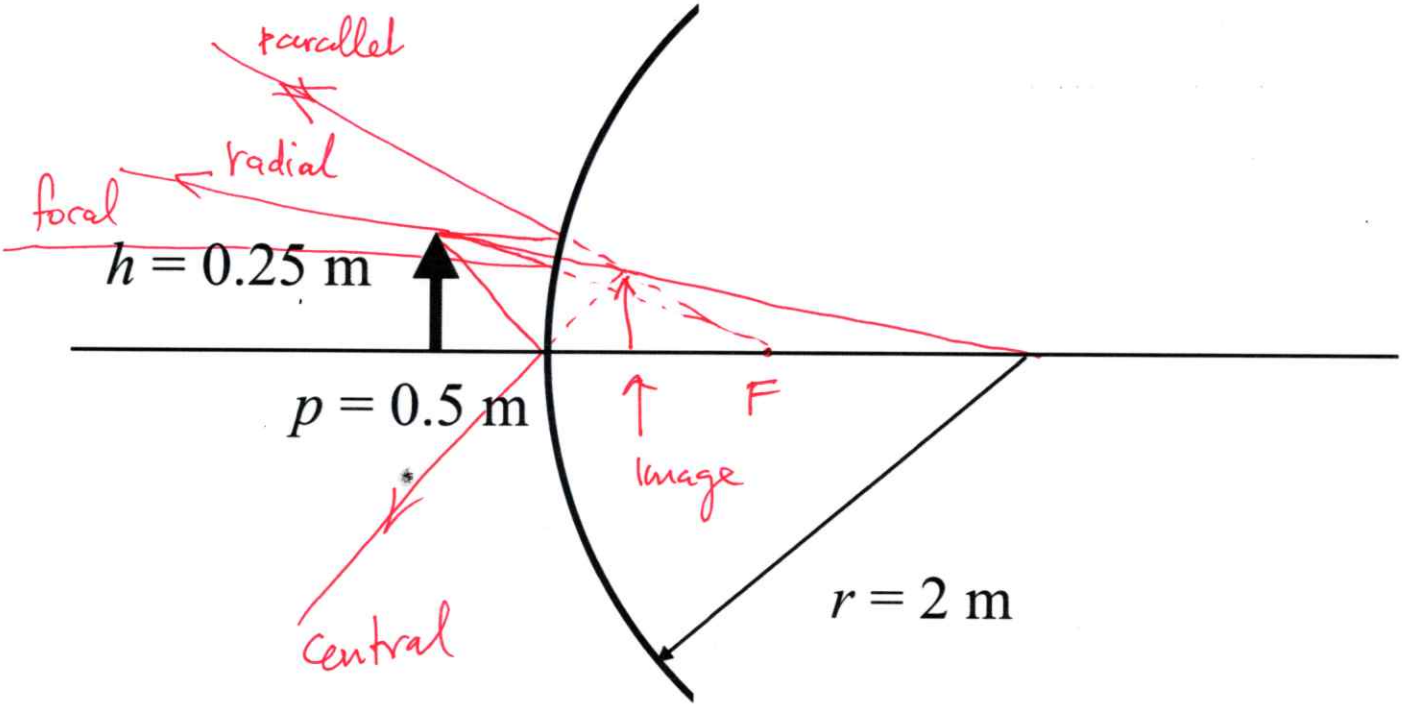
(e) (4 points) Check your answers with a ray diagram in the space below (use at least two rays, label them or explain in words how to draw them. The third ray is a bonus 1 point)



+2 for each ray

+1 ← bonus for the correct third

19. (12 points) A 0.25m tall arrow is placed at the indicated location in front of a spherical mirror with properties shown in the figure below.



(a) (2 points) Determine the focal length of the mirror, including the sign.

$F = -\frac{r}{2} = -1\,m$, "–" because the mirror is convex

whether

(b) (4 points) Using ray tracing methods that employ at least two rays (any extra ray gets +1 bonus point), determine wethen the image formed by the mirror is real or virtual, inverted or non-inverted, and magnified or demagnified. Label the rays or explain how they are plotted.

(c) (6 points) Using algebraic methods, determine the location and size of the image. Compare your results with those obtained by the ray tracing methods.

+2 $\frac{1}{p} + \frac{1}{i} = \frac{1}{f} \Rightarrow$

$\frac{1}{i} = \frac{1}{f} - \frac{1}{p} = -\frac{1}{1m} - \frac{1}{0.5m} = -3\,m^{-1}$

+2 $i = -0.33$ m ← behind mirror

$m = -\frac{i}{p} = -\frac{-0.33\,m}{0.5\,m} = 0.66$ $h' = mh = 0.25m \cdot 0.66 \approx 0.17m$

+2

20. (10 points) You are given Young's double slit experiment with a separation between the slits $d = 0.2$ mm, width of each slit $a = 0.04$ mm, and a source of coherent light with the wavelength $\lambda = 500$ nm.

(a) (3 points) Determine the angular separation between individual interference maxima on a screen produced by this double-slit setup.

(+1) $d \sin\theta = n\lambda \Rightarrow \sin\theta = \frac{n\lambda}{d}$ or $\theta_n \simeq \frac{n\lambda}{d}$

$\theta_{n+1} - \theta_n = \frac{n+1}{d}\lambda - \frac{n\lambda}{d} = \frac{\lambda}{d} = \frac{500\,nm}{0.2\,mm} \simeq 0.0025\,rad$

or $\sin\theta_1 = \frac{\lambda}{d} \Rightarrow \theta_1 \simeq a\sin\left(\frac{\lambda}{d}\right) \simeq 0.0025\,rad \simeq 0.29^\circ$ (+2)

(b) (3 points) Determine the angle between the center and the first diffraction minimum.

(+1) $a \sin\theta = m\lambda, \quad m = \pm 1, \pm 2 \ldots$

$\sin\theta_1 = \frac{\lambda}{a} = \frac{500 \cdot 10^{-9}\,m}{4 \cdot 10^{-5}\,m} \simeq 0.0125 \Rightarrow \theta_1 \simeq 1.25 \cdot 10^{-2}\,rad$

$\simeq 0.72^\circ$

(+2)

(c) (2 points) How many interference maxima are located between the center and the first diffraction minimum?

$\# = \frac{1.25 \cdot 10^{-2}\,rad}{0.0025\,rad} = 5$ (or 6 depending how one counts)

(d) (2 points) Sketch a light pattern that you would expect to observe on the screen produced by this double-slit experiment.

Some effort to draw interference & diffraction pattern (+2)